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Building Web Clipping Applications with HTML

Scot Stennis
Palm Development Support Engineer

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Message

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- Developing web clipping applications is easy and fast

Agenda

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- Web Clipping Applications
- What You Need
- 7 Easy Steps
- Design Guidelines
- The Details
- Advanced Topics
- Development Resources
- Conclusion
- Q&A

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Web Clipping Applications

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Web Clipping Applications

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- What are they?
  - Palm™ database files
    - Pages and images are records
- How are they used?
  - Like a web page
    - Tap on the links
    - Fill out and submit the forms
  - Rendered by the WCA Viewer or “Clipper”
    - Transparent to the user
- What do they offer?
  - Simple wireless Internet access for a hand-held device


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What You Need
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What You Need

Tools

- Text editor
  - e.g., Notepad or BBEdit
- Graphics program
- WCA Builder
  - Windows® or Macintosh® platform
- Palm OS® Emulator (POSE)
  - Palm OS 3.2 ROM file
- Palm™ Desktop Software
- HotSync® cradle and Palm VII™ device

What You Need

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Skills

- HTML
- CGI scripting
- Others
  - Web server configuration
  - Firewall configuration

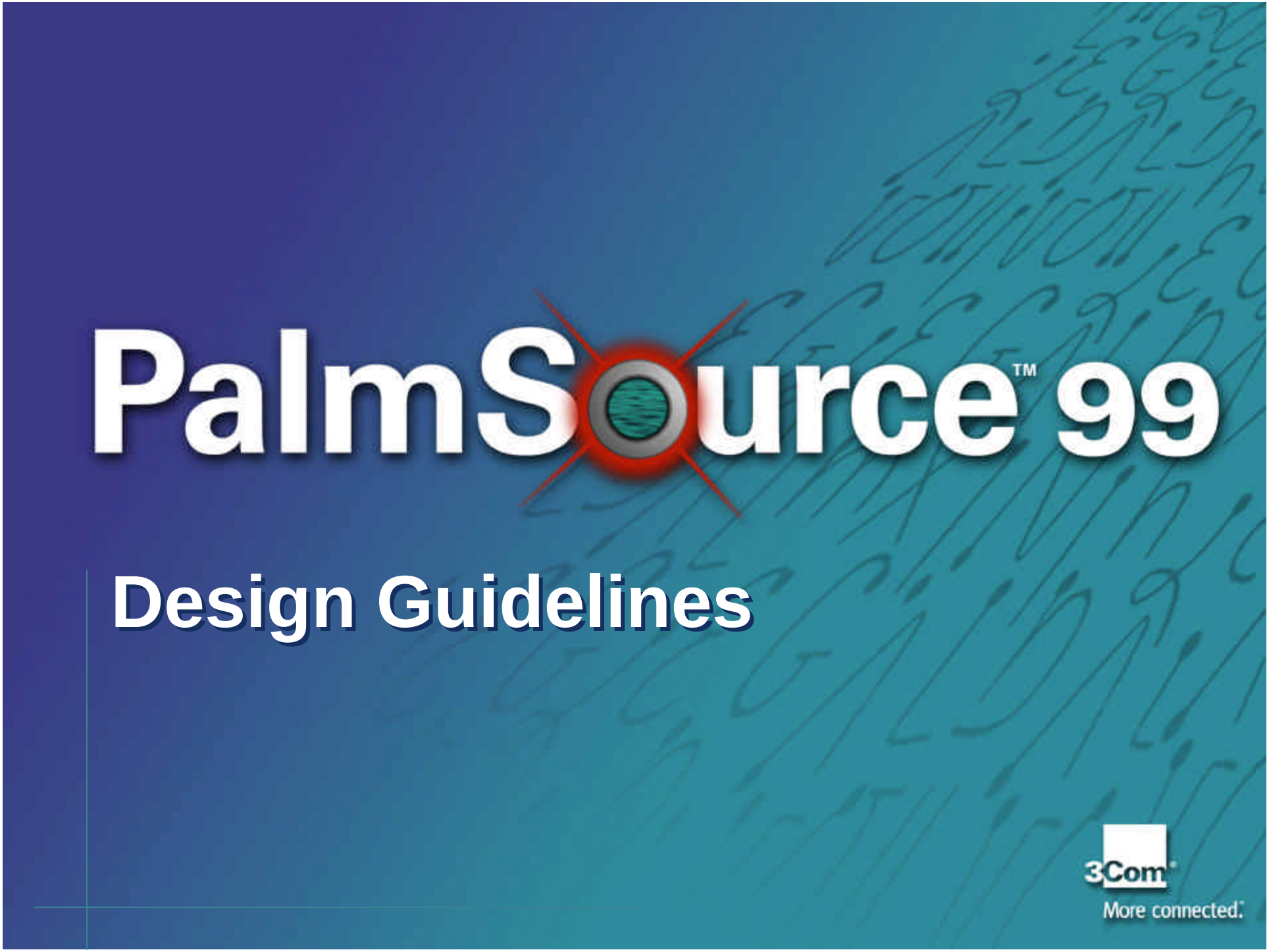

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7 Easy Steps
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7 Easy Steps

1. Design
2. Code in HTML
3. Convert the HTML into a WCA
4. Install the WCA
   - Palm OS® Emulator
   - Palm VII™ device
5. Test
6. Debug



2. Code back-end server


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Design Guidelines

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- Establish target audience
- Determine content
  - Key elements
  - Client (WCA)
  - Server (CGI, servlets)
- Think small
  - Screen size (156 x 160)
  - Bytes per request and response
  - Tables
  - Avoid need for scrolling
    - Main page
    - Clippings

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The Details

HTML Set Overview

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HTML Set Overview

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- Meta tags
  - PalmComputingPlatform
  - PalmLauncherRevision
  - LocalIcon
  - HistoryListText
- Form input element types
  - DatePicker
  - TimePicker

HTML Set Overview

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- Special strings
  - %deviceid
  - %zipcode
- SmallScreenIgnore
- Referencing
- Graphics
- Unsupported listing

HTML Set Overview

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Meta Tags: PalmComputingPlatform

- Palm-friendly indicator
- Needed by WCA files
- Needed by clippings

```
<meta name="PalmComputingPlatform"
      content="true">
```

HTML Set Overview

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Meta Tags: PalmLauncherRevision

- Version number for the PQA
  - Seen in the Applications Launcher screen, version info view

```
<meta name="PalmLauncherRevision"
      content="1.0">
```

HTML Set Overview

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Meta Tags: LocalIcon

- Used to include images or html pages not referenced locally
- Placed in PQA, not clippings

```
<meta name="LocalIcon"
      content="image.gif">
```

HTML Set Overview

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Meta Tags: HistoryListText

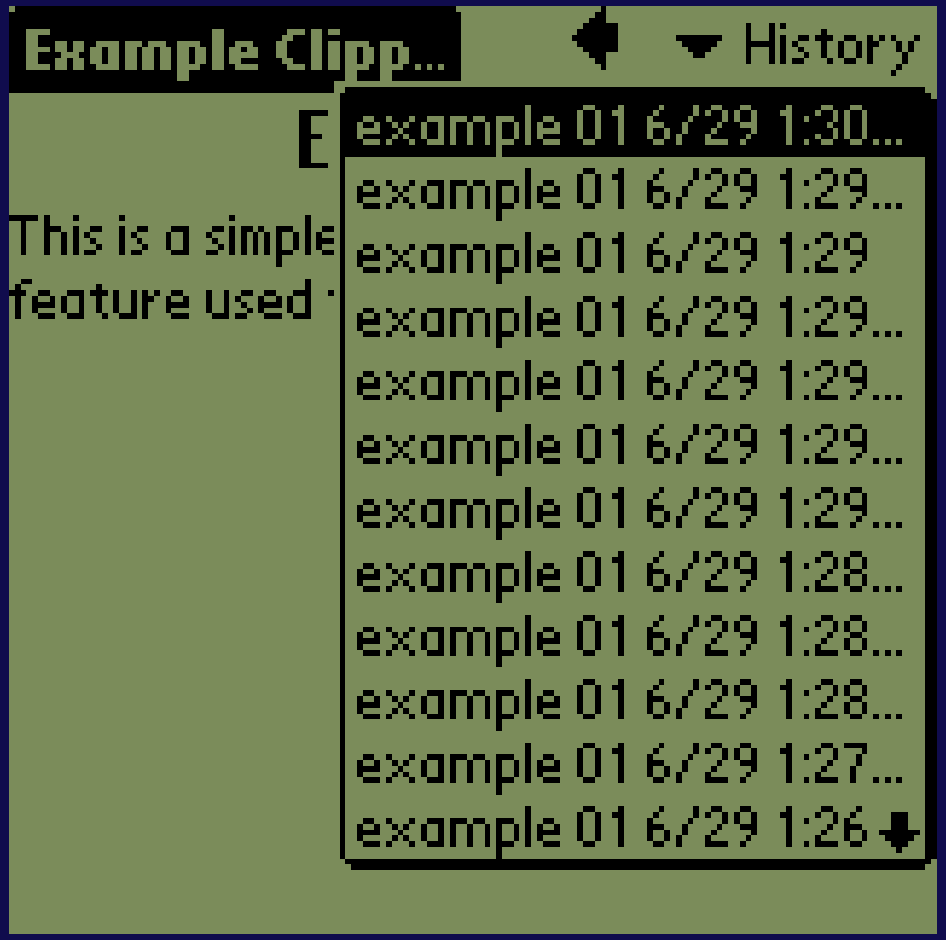
- Provides informative titles for cached pages
- &date and &time strings in the content attribute

```
<meta name="HistoryListText"
      content="string to display &date &time">
```

HTML Set Overview

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Meta Tags: Example Screen



HTML Set Overview

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Form Input Element Types: DatePicker

- Displays as set, like MM-DD-YY
- Displays the current date if value attribute is undefined
- Sends YYYY-MM-DD format

```
<input type="datepicker"
       name="date"
       value="">
```

HTML Set Overview

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Form Input Element Types: TimePicker

- Displays as set, like HH:MM am/pm
- Displays the current time if the value attribute is undefined
- Sends HH:MM 24-hour clock format

```
<input type="timepicker"
       name="time"
       value="">
```

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Special Strings: %DeviceId

- Two part format:
  - Validation code
    - -1 = invalid device id
    - 0 = couldn't validate id
    - 1 = valid device id
  - Device identifier
    - Maximum of 64 bytes
    - Black box—may change in the future
- Not for use in clippings
  - Replaced by device with a series of characters
  - Mapped by proxy server to numeric id

HTML Set Overview

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Special Strings: %ZipCode

- Often used for auto-retrieve features
  - Local news, weather, movies, etc.
- Not for use in clippings
  - Clipping proxy server replaces with the zipcode of the local base station

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Special Strings: Examples

```
<form method="get"
      action="http://www.x.com/cgi/x.cgi">
<input type="hidden"
       name="did"
       value="%deviceid">
<input type="hidden"
       name="zip"
       value="%zipcode">
…</form>

<a href="http://www.x.com/x.cgi?did=%deviceid">Go</a>
<a href="http://www.x.com/x.cgi?zip=%zipcode">Go</a>
```

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SmallScreenIgnore

- An element used to contain HTML that you don’t want sent over-the-air to a device
  - Content is removed by the clipping proxy server

```
<smallscreenignore>
...
<img src=“reallyBig.gif” height=“200” width=“300”>
...
</smallscreenignore>
```

HTML Set Overview

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Referencing

- Referencing WCA pages from WCA pages:

```
<a href="about.html">About</a>
```

or

```
<a href="about.html" button>About</a>
```

or

```
<img src="cool.gif">
```

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Referencing

- Referencing WCA pages from clippings:

```
<a href="file:mypqa.pqa/about.html">About</a>
```

or

```
<img src="file:mypqa.pqa/cool.gif">
```

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Referencing

- Referencing the Internet:

  <a href="http://www.palm.com">

  - Over-the-air symbol automatically included

  <a href="https://www.bofa.com">

  - Secure over-the-air symbol automatically included

  - Over-the-air symbols not included after linked graphics

HTML Set Overview

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Graphics

- Supported formats
  - gif, jpg, bmp
- 1 or 2-bit grayscale
- 32x22 pixels for large launcher icon
- 15x9 pixels for small launcher icon
- 153 pixel width max for referenced images
- Sequential gif animation is not supported

HTML Set Overview

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Unsupported Listing

- Applets
- JavaScript
- Frames
- Nested tables
- Image maps
- Cookies
- VSPACE attribute
- <SUB>
- <SUP>
- <LINK>
- <ISINDEX>

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The Details

Back-end Servers

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Back-end Servers

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- HTTP or HTTPS
  - Don’t require a certificate from the clipping proxy server when using HTTPS and SSL
- Requests
  - HTML content
  - %deviceid and %zipcode values
- Clippings
  - HTML content
  - Same HTML set as for the WCAs
  - Don’t use %deviceid nor %zipcode in clippings

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The Details

Common Pitfalls

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Common Pitfalls

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Web Clipping Applications

- Forgot to include the PalmComputingPlatform meta tag
- Referenced using the file:mine.pqa/about.html syntax from the root page

Common Pitfalls

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Web Clippings

- Forgot to include the PalmComputingPlatform meta tag
- Attempted to re-direct to a page stored on the device
- Used relative referencing to pages stored on the device
- Sent tables wider than 156 pixels

Common Pitfalls

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Back-end Servers

- Forgot to account for the Palm OS® Emulator deviceid of 0.0.0
- Requiring a certificate from the clipping proxy server when establishing an SSL connection

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Advanced Topics

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Advanced Topics

- Palm
  - Used to quit Clipper and call another application
- Palmcall
  - Similar to palm, but returns control to Clipper
  - Application called needs to:
    - Parse arguments
    - Have means for user to quit
- Palm VII™ SDK
  - INetLib API
    - Provide URL, get response
    - Similar to Berkley Sockets functions


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Development Resources

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- www.palm.com/devzone/webclipping
  - Web Clipping Developer's Guide
  - WCA Builder
  - Palm OS® Emulator
  - Tutorials
  - Example HTML code
- http://www.palm.com/devzone/assistance/
  - Mailing lists
  - Training info
  - Knowledge base
  - Newsgroup listings
  - Submit a technical question

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Conclusion

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Conclusion

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Summary

- WCAs are like a miniature web site
- WCAs are easy to build
  - Text editor
  - Web Clipping Application Builder
  - Palm OS® Emulator
  - Palm OS™ 3.2 ROM file
  - Palm VII™ Connected Organizer
- Think small
- Subset of HTML 3.2 used

Conclusion

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What Next?

- Session 448
  - Learn about palm and palmcall
  - Learn how to quickly add wireless capability to your .prc
- Session 480 Lab
  - Learn how to use the tools for building WCAs
  - Get hands-on use of the tools
  - Build an example WCA
- Session 450
  - Learn about the INetLib API

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Q&A

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